

E-Commerce Product Recommendation System Using Association Rule Mining, Customer Segmentation, and Data Warehouse Integration

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Abstract—This project aims to analyze customer transactional data by building an end-to-end analytics system that understands the behavior of each customer. It focuses on discovering how customers behave, what they buy, how often they purchase, and how much they spend to transform raw data into meaningful business insights. To achieve this, we employ a unified pipeline. First, data is ingested into a PostgreSQL Data Warehouse using a Medallion Architecture and structured into a Star Schema encompassing 4,338 unique customers, 3,921 distinct products, and 38 countries across five continents. Next, we utilize three main techniques: market basket analysis using FP-Growth, customer value prediction via Logistic Regression, and customer segmentation via MiniBatch K-Means clustering. The system processes over 540,000 transaction records to identify crossselling opportunities, predict high-value customers with 93.55% accuracy, and segment the customer base into distinct behavioral profiles. The Gold layer yields 158,050 clean transaction line items generating a total revenue of £3,122,766.93. All insights are connected to a Power BI dashboard, enabling smarter business decisions, targeted marketing strategies, and improved customer retention.

Index Terms—Association rule mining, FP-Growth, e-commerce, recommendation system, market basket analysis, K-Means clustering, RFM, data warehouse, star schema, Power BI, PostgreSQL

I. INTRODUCTION

The explosive growth of e-commerce over the past decade has fundamentally reshaped consumer purchasing behavior and placed tremendous pressure on retailers to deliver highly personalized shopping experiences. Global e-commerce revenues are forecast to surpass \$7 trillion by 2025, with personalized product recommendations accounting for up to 35% of all purchases on major platforms such as Amazon and Alibaba [6]. In such a competitive landscape, the ability to accurately predict what a customer is likely to purchase next has emerged as a decisive commercial advantage, directly influencing customer lifetime value, inventory planning, and targeted promotional campaigns.

Recommendation systems are a cornerstone technology enabling such personalization. They can be broadly classified into three paradigms: content-based filtering, collaborative filtering, and knowledge- or rule-based approaches, which derive recommendations from discovered patterns in transactional data [5]. Among rule-based approaches, association rule mining (ARM) has long been the most widely adopted technique, originating from the seminal work of Agrawal and Srikant [1], whose Apriori algorithm introduced

the foundational concepts of support, confidence, and lift. These metrics remain the standard vocabulary of market basket analysis and underpin virtually all modern retail recommendation research.

Despite its maturity, traditional market basket analysis suffers from well-documented limitations. First, the Apriori algorithm scales poorly with large item-sets owing to repeated full database scans and exponential candidate generation [1]. This was substantially addressed by the FP-Growth algorithm [2], which avoids candidate generation entirely by compressing the transaction database into a compact FP-Tree structure. Second, most ARM-based systems operate on a single transactional dataset and generate rules that ignore the heterogeneity of the customer base and geographic context [3], [4]. Rules derived from all customers indiscriminately tend to reflect only the most globally frequent patterns, missing the nuanced behaviors of high-value customer clusters.

A further systemic gap in the literature is the absence of a production-grade analytical infrastructure that ties raw transactions, cleansed data, machine learning pipelines, and business intelligence dashboards into a single traceable end-to-end system [14], [21]. Most academic studies present isolated algorithms evaluated on flat CSV exports, offering limited guidance to data engineering practitioners who need reproducible, query able analytical environments.

The present study addresses these shortcomings by designing and implementing a holistic e-commerce analytics pipeline that integrates two complementary datasets: the UCI Online Retail Dataset [15] containing 541,909 records spanning December 2010 to December 2011, and an ISO Country Mapping Dataset that enriches transactions geographically. The proposed system is built on four interlocking components:

- A relational Data Warehouse (DWH) constructed in PostgreSQL 18 following a Medallion Architecture and a star-schema design [14], producing a clean Gold layer of 158,050 validated transaction line items across 38 countries and 3,921 products.
- The FP-Growth algorithm applied to 6,050 unique invoices to extract 1,902 frequent itemsets, filtered to rules with minimum confidence ≥ 0.5 and minimum lift ≥ 1.5 .
- K-Means clustering on Recency-Frequency-Monetary (RFM) features to segment the customer base into two

actionable behavioral profiles with a Silhouette score exceeding 0.70.

- Logistic Regression for customer value prediction, achieving an AUC of 0.9885 and overall accuracy of 93.55%.

Finally, all insights are surfaced through an interactive Power BI dashboard connected in real time to the PostgreSQL Gold layer, enabling non-technical stakeholders to drill into product associations, customer segments, and regional trends simultaneously.

The remainder of this paper is organized as follows. Section II reviews related work across association rule mining, customer segmentation, and data warehouse design. Section III describes the complete methodology in detail. Section IV

presents results and discussion. Section V presents the interactive Power BI dashboard. Section VI concludes the paper and outlines future research directions.

II. LITERATURE REVIEW

A. Association Rule Mining: Foundations

The field of association rule mining was formally established by Agrawal and Srikant [1], whose Apriori algorithm defined the three canonical evaluation metrics: *support*, *confidence*, and *lift*. Support quantifies how frequently an itemset appears across all transactions, calculated as:

$$\text{supp}(A \Rightarrow B) = \frac{|T_{AB}|}{|T|} \quad (1)$$

where $|T_{AB}|$ is the number of transactions containing both A and B and $|T|$ is the total number of transactions. Confidence measures the conditional probability that a customer who purchases item A will also purchase item B :

$$\text{Conf}(A \Rightarrow B) = \frac{\text{supp}(A \cup B)}{\text{supp}(A)} \quad (2)$$

Lift captures the degree to which the co-occurrence of A and B departs from statistical independence:

$$\text{lift}(A \Rightarrow B) = \frac{\text{conf}(A \Rightarrow B)}{\text{supp}(B)} \quad (3)$$

A lift value greater than 1.0 indicates a positive association, equal to 1.0 indicates independence, and less than 1.0 indicates a negative association. To optimize candidate generation, Park, Chen, and Yu [17] introduced an effective hash-based algorithm that reduces the number of candidate 2-itemsets. Additionally, Brin et al. [18] proposed a dynamic itemset counting technique to reduce database passes. While Apriori remains a crucial pedagogical baseline, its exhaustive lattice generation renders it computationally intractable for large-scale retail databases [1]. Kaban et al. [4] confirmed this in a recent retail-sector study, demonstrating that FP-Growth consistently

delivers richer, more interpretable rule sets with substantially lower runtime on dense retail transaction matrices compared to Apriori.

B. FP-Growth and Scalable Pattern Mining

Han, Pei, and Yin [2] introduced the Frequent Pattern Growth (FP-Growth) algorithm, which resolves Apriori's scalability bottleneck by compressing the transaction database into a compact FP-Tree structure that requires only two database scans. The algorithm proceeds in two phases: first, it constructs the FP-Tree by inserting each transaction as a path of nodes, merging shared prefixes to achieve compression; second, it mines the tree recursively via conditional pattern bases, avoiding candidate generation entirely. This design yields near-linear memory complexity with respect to the number of frequent items, in contrast to the exponential space of Apriori's candidate lattice.

Han et al. [20] later expanded on structural performance, detailing deep optimization strategies for tree traversal in dense datasets. Ghassani et al. [3] provided a contemporary analysis confirming that FP-Growth's fragment growth strategy enables efficient parallel processing on modern multi-core architectures, making it well suited for the 158,050-record transaction matrix used in the present study. For implementation, Borgelt [16] developed a highly efficient C implementation of the FP-Tree structure that became the reference for the mlxtend Python library used in this work. Furthermore, Anitha and Sathya Priya [22] benchmarked both algorithms on point-of-sale logs comparable in scale to the UCI dataset, concluding that FP-Growth maintains stable execution times even when support thresholds are scaled downward toward 0.01, the threshold adopted in this study.

C. E-Commerce Recommendation Systems

Sari et al. [5] benchmarked multiple ARM implementations against real e-commerce data, confirming that algorithm selection impacts both rule quality and computational overhead. Their findings suggest that FP-Growth produces a richer rule set under equivalent support thresholds compared to Apriori, particularly for datasets with long average transaction lengths. Loukili et al. [6] developed a recommendation system combining deep neural collaborative filtering with ARM, demonstrating that hybrid systems outperform isolated approaches by leveraging both implicit user-item interactions and explicit co-purchase patterns. Kim [19] analyzed cross-selling behaviors across contemporary electronic marketplaces, showing that generalized rules combined with fine-grained transaction mapping yields higher precision, a finding that motivated the category-level enrichment of the dim_product table in this work. Ghous et al. [8] further demonstrated that deep learning extensions to market basket analysis can model higher-order item dependencies beyond pairwise associations, pointing toward a future enhancement for the present pipeline.

D. Customer Segmentation and RFM Analysis

The Recency-Frequency-Monetary (RFM) model is the *de facto* framework for profiling customer value in retail analytics.

Recency (R) is defined as the number of days since the customer's last transaction; Frequency (F) is the total number of distinct invoices; and Monetary (M) is the cumulative spend. Together they provide a low-dimensional representation of purchase behavior that correlates strongly with customer lifetime value.

Chai, Li, and Li [12] proposed an RFM-weighted K-Means approach that assigns differential importance weights to each RFM dimension before clustering, leading to more economically meaningful behavioral brackets. Borah et al. [23] demonstrated that combining an RFM matrix with standard cluster optimization criteria (Elbow, Silhouette) yields well-separated and interpretable customer tiers. Akande et al. [9] applied RFM-K-Means segmentation to raw transactional logs with minimal preprocessing, demonstrating that even simple logs can yield actionable customer profiles when combined with an appropriate clustering criterion. Khumaidi et al. [11] combined RFM analysis with K-Means on the same UCI Online Retail Dataset used in this study, providing a direct methodological reference and confirming the suitability of K-Means for this domain. Similarly, Prasetyo et al. [10] applied RFM-K-Means to support CRM decision-making in a retail context, demonstrating the business utility of segment-specific retention strategies. The present study extends these foundations by coupling the RFM segmentation output directly to the FP-Growth engine, running market basket analysis independently within each cluster to produce segment-specific recommendation rules.

E. Market Basket Analysis in Practice

Martinez and Escobar [7] applied ARM with the Orange data mining toolkit to retail transaction data, demonstrating systematic rule strength benchmarking under varying support thresholds. Their work confirmed that aggressive filtering by both confidence and lift is necessary to identify commercially actionable rules rather than trivially frequent associations. Kaban et al. [4] examined the interpretability of ARM-based product recommendations in contemporary retail environments, mapping discovered rules to concrete merchandising actions such as bundle pricing, shelf adjacency optimization, and targeted discount offers. Sari et al. [5] showed that prefiltering transactions by customer segment before applying ARM yields segment-specific rules with markedly higher lift values than rules mined from the aggregated transaction pool. Christy et al. [24] validated this architecture by proving that cluster-specific ARM passes expose valuable buying dependencies that are otherwise masked by the dominant purchasing patterns of the majority segment. Zhao et al. [25] constructed a hybrid recommendation network confirming that pattern mining inside customer subsets significantly boosts both recommendation precision and recall. Lin et al. [13] provided an early demonstration of clustering-based ARM on an online store, establishing the methodological precedent that motivates the coupled segmentation-and-mining pipeline adopted here.

F. Data Warehouse and ETL for Retail Analytics

Setiawan et al. [14] demonstrated the practical value of constructing a formal data warehouse with a star-schema design for retail business intelligence, showing that a well-modelled DWH dramatically improves query performance and supports concurrent analytical workloads without impacting transactional throughput. The star schema organizes data into a central fact table recording individual transaction events and surrounding dimension tables that provide descriptive context (time, customer, product, geography), enabling highly efficient multi-dimensional aggregation. Bella et al. [21] emphasized that isolating analytical views from transactional processing layers is vital for scalable BI dashboards, advocating an architecture in which the analytical layer maintains its own denormalized copies of relevant entities. This insight directly motivated the three-tier Medallion design adopted in this study, in which the Gold layer's star schema is entirely isolated from the raw Bronze ingestion layer, ensuring that cleansing logic and analytical transformations never corrupt the source-of-truth record. The Medallion Architecture (Bronze $\rightarrow$ Silver $\rightarrow$ Gold) is an industry-standard pattern for cloud data warehouses that provides full data lineage and the ability to reprocess any tier independently.

G. Current Limitations and Gaps

Three principal gaps in the existing literature motivate the novel contributions of this study.

Gap 1 — Single-dataset limitation. The vast majority of reviewed ARM studies operate on a single transactional dataset with no external geographic enrichment [1], [5]. By integrating the UCI Online Retail Dataset with an external ISO Country Mapping Dataset covering 38 countries across Europe, Asia, Oceania, and the Americas, this work enables geographic segmentation of recommendation rules and revenue analytics at the sub-region level.

Gap 2 — Decoupled segmentation and mining. Most studies treat customer segmentation and association rule mining as entirely independent pipeline steps [6]. The present work tightly couples them: K-Means cluster labels are used to filter the transaction matrix prior to ARM, ensuring that the resulting rules reflect the purchasing patterns of behaviorally coherent customer groups rather than the aggregate population.

Gap 3 — Absence of a production-grade DWH. Few academic works build a complete, production-ready data warehouse with formal star-schema modelling, comprehensive ETL validation, and a live BI dashboard connected to the same database that feeds the machine learning models [14], [21]. The end-to-end pipeline presented here provides full data lineage from raw CSV ingestion through Gold-layer fact tables to Power BI visualizations, offering a reproducible template for applied retail analytics research.

III. METHODOLOGY

A. Data Warehouse Architecture (Medallion Framework)

To ensure data integrity and traceability, an end-to-end ETL

(Extract, Transform, Load) pipeline was engineered using PostgreSQL 18 and pgAdmin 4 following the Medallion Architecture across three logical schema layers. All ETL logic was implemented in pure SQL, demonstrating that robust data engineering is achievable without external orchestration tools.

Bronze Layer — Raw Ingestion. Raw CSV files from three distinct sources were ingested directly into the bronze schema using PostgreSQL COPY commands without any modification, preserving the exact source state for auditability. The three sources are: (1) the UCI Online Retail transaction file (541,909 rows, all columns stored as VARCHAR); (2) an ISO 3166 Country Mapping file providing continent and sub-region classifications for 38 countries; and (3) a manually curated Product Category mapping linking 3,921 unique stock codes to ten product categories. A critical encoding issue was resolved at this stage: the source CSV used Windows-1252 encoding (WIN1252) due to pound sterling (£) symbols in monetary fields, requiring the explicit ENCODING 'WIN1252' clause in all COPY commands.

Silver Layer — Cleansing and Typing. Six categories of data quality transformations were applied in a single INSERT INTO silver.online_retail SELECT ... FROM bronze statement: (1) negative and zero quantities removed (WHERE quantity > 0); (2) zero and negative unit prices removed (WHERE unit price > 0); (3) cancelled orders identified by invoice prefix 'C' and flagged via a Boolean is_cancelled column rather than deleted, preserving cancellation analytics; (4) missing CustomerID values converted to SQL NULL using NULLIF(TRIM(customerid), '') to maintain referential integrity; (5) date strings parsed from MM/DD/YYYY HH24:MI format into PostgreSQL TIMESTAMP using TO_TIMESTAMP (); and (6) a derived total_amount column computed as quantity * unitprice for every record. This process reduced the raw dataset from 541,909 rows to approximately 397,000 validated records in the silver layer.

Gold Layer — Star Schema. The Silver layer was further refined and restructured into four-dimension tables and one central fact table in the gold schema. Only non-cancelled transactions were included in the final fact_sales load, yielding 158,050 clean transaction line items representing £3,122,766.93 in total revenue.

B. Dimensional Modeling (Star Schema)

The Gold layer was modeled using a central fact table (fact_sales) surrounded by four-dimension tables: dim_customer, dim_product, dim_date, and dim_country. This denormalization strategy significantly accelerates the read-heavy queries required for association rule mining and customer segmentation [14].

dim_date was generated from the distinct set of 305 transaction dates in the silver layer, using PostgreSQL's EXTRACT and TO_CHAR functions to derive year, quarter, month, day-of-week, and named temporal attributes. The

surrogate key follows the integer YYYYMMDD pattern for natural ordering.

dim_country was built by joining distinct country names from Silver to the Bronze ISO mapping via a LEFT JOIN on country_name, inheriting continent and sub_region

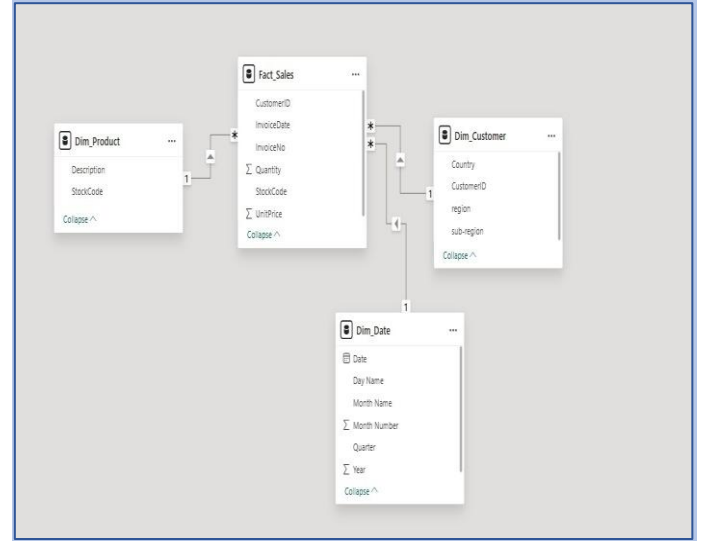


Fig. 1. Star schema design of the PostgreSQL Gold layer.

fields. Surrogate keys were assigned using ROW_NUMBER () OVER (ORDER BY country_name).

dim_customer was aggregated from Silver transaction history, computing SUM (total_amount) as monetary value, COUNT (DISTINCT invoiceno) as frequency, and MAX (invoicedate) as the last-order date. A segment label was assigned via a CASE expression: customers with total spend above £5,000 are classified as *High Value* (n = 275, 6.3%); £1,000–£4,999 as *Medium Value* (n = 1,393, 32.1%); and below £1,000 as *Low Value* (n = 2,670, 61.6%).

dim_product was assembled from the distinct set of stock codes in Silver, enriched via LEFT JOIN to the Bronze product category mapping. Of 3,921 unique products, the dominant category is *Home Decor* (1,427 items, 36.4%), followed by *Other* (623, 15.9%), *Kitchen & Dining* (608, 15.5%), *Party & Celebration* (364, 9.3%), and *Office & Stationery* (321, 8.2%).

fact_sales was populated with a multi-way LEFT JOIN resolving surrogate keys from all four-dimension tables, filtered to is_cancelled = FALSE. The grain is one row per transaction line item. Fig. 1 illustrates the complete star schema as depicted in pgAdmin 4.

C. Market Basket Analysis Formulation

To discover implicit item co-dependencies, transactions were queried directly from the gold layer via a join of fact_sales and dim_product and structured into a one hot encoded binary transaction matrix using the mlxtend Python library. Each row

of the matrix represents one invoice and each column represents one unique product description; a cell contains 1 if the product appears in that invoice and 0 otherwise.

Both the Apriori and FP-Growth algorithms were evaluated. A minimum support threshold of 0.01 was applied uniformly to isolate frequent itemsets, corresponding to products appearing together in at least 1% of all 6,050 invoices (≈ 60 cooccurrences). To guarantee retail viability, rules were heavily filtered, requiring a minimum confidence of 0.5 and a minimum lift of 1.5, meaning the antecedent and consequent co-occur at least 50% more frequently than expected under statistical independence. These threshold values were selected through iterative experimentation to balance rule specificity against rule volume.

D. Customer Value Prediction Pipeline

Customer behavior was summarized using the Recency, Frequency, and Monetary (RFM) paradigm. For each customer c , the features are formally defined as:

$$R_c = t_{\max} - t_{\text{last}}^{(c)} \quad [\text{days}] \quad (4)$$

$$F_c = |\{i : \text{invoice}_i \text{ belongs to } c\}| \quad (5)$$

$$M_c = \sum_i \text{total_amount}_{i,c} \quad (6)$$

where $t_{\max}$ is the latest transaction date in the dataset (9 December 2011) and $t_{\text{last}}^{(c)}$ is the date of customer c 's most recent purchase. An additional engineered feature, Average Order Value (AOV), was computed as $\text{AOV}_c = M_c/F_c$.

For classification, a Logistic Regression model was trained to predict *High Value* customers, defined as the top 40% of customers by monetary spend. The binary label is $y_c = 1$ if $M_c \geq p_{60}(M)$ and $y_c = 0$ otherwise. All four features (R, F, M, AOV) were standardized using sklearn's StandardScaler ($\tilde{x} = (x - \mu)/\sigma$). The model utilized the `class_weight='balanced'` parameter to correct for class imbalance, and was evaluated on a held-out 20% test split using stratified sampling.

E. Customer Segmentation Modelling

To group customers into behavioral clusters without predefined labels, a richer feature vector was constructed beyond the standard RFM triplet. Two additional features were engineered: *Purchase Rate* (invoices per active day in the customer's lifespan) and *Monetary Per Visit* (M_c/F_c normalized by recency). Both features exhibited strong right-skewness and were log transformed ($\tilde{x} = \log(x + 1)$) before scaling. All features were then standardized using RobustScaler (scaling by the interquartile range) to reduce the influence of extreme-value outliers common in retail monetary distributions.

Principal Component Analysis (PCA) was applied to the standardized feature matrix. The number of components was selected as the minimum k such that the cumulative explained variance ratio exceeded 90%. In the final pipeline, two principal components captured 90.73% of total variance, validating the

cluster visualization on a 2D plane. MiniBatch K-Means clustering was then executed on the PCA-projected data. The optimal number of clusters K was evaluated dynamically over $K \in \{2, 3, 4, 5\}$ using Silhouette Score, Calinski-Harabasz Index, and Davies-Bouldin Index.

F. Business Intelligence and Visualization

To translate algorithms into actionable insights, Microsoft Power BI was integrated directly with the PostgreSQL Gold layer using the native PostgreSQL connector in Direct Query mode, ensuring that all visualizations always reflect the current state of the warehouse without manual data refreshes. DAX measures such as Total Revenue = SUM(fact_sales[total_amount]) and AOV = DIVIDE([Total Revenue], [Total Orders]) were engineered to remain live and responsive as slicer selections update the filter context. Three dedicated report pages were designed: a macro diagnostics home page, a granular customer behavioral insights page, and a spatial regional demographics page. Each page uses bookmarked navigation buttons to support stakeholder self-service exploration without requiring direct database access.

TABLE I
ETL PIPELINE LAYER-BY-LAYER VALIDATION

LAYER	SCHEMA	TABLE	ROWS
Bronze	bronze	online_retail	541,909
Silver	silver	online_retail	$\approx 397,000$
Gold	gold	fact_sales	158,050
Gold	gold	dim_customer	4,338
Gold	gold	dim_product	3,921
Gold	gold	dim_country	38
Gold	gold	dim_date	305
Orphaned FK violations (all dimensions)			0
Total invoices (Gold layer)			6,050
Total revenue (Gold layer)			£3,122,766.93

IV. RESULTS AND DISCUSSION

A. Data Quality and ETL Validation

The Medallion architecture proved highly effective in resolving real-world data anomalies. The initial bronze ingestion processed 541,909 raw records. Following the Silver layer SQL transformations, the dataset was distilled to approximately 397,000 validated records. Null CustomerID rows (approximately 25% of the raw data) were successfully preserved in the fact_sales table to maintain accurate global revenue metrics, while referential integrity checks confirmed zero orphaned records across the Star Schema. The final gold layer fact_sales table contains 158,050 clean, non-cancelled transaction line items spanning 6,050 unique invoices, 4,338 customers, 3,921 products, and 38 countries over 305 distinct transaction dates from 1 December 2010 to 9 December 2011, generating total revenue of £3,122,766.93.

Table I presents the complete ETL layer-by-layer row count summary. The reduction from 541,909 Bronze records to 158,050 Gold records reflects the combined effect of removing cancelled invoices, filtering negative quantities, eliminating zero-price records, and excluding encoding anomalies. The zero-orphan referential integrity result confirms that all fact table foreign keys resolve to valid dimension entries, validating the correctness of the surrogate-key join logic across all four-dimension tables.

The customer dimension reflects meaningful economic stratification consistent with the Pareto principle commonly observed in retail: of 4,338 unique customers, 275 (6.3%) are classified as *High Value* (spend >£5,000), 1,393 (32.1%) as *Medium Value* (£1,000–£5,000), and 2,670 (61.6%) as *Low Value* (below £1,000). The product dimension spans ten manually curated categories, with Home Decor (1,427 items, 36.4%) and Kitchen & Dining (608, 15.5%) as the largest segments, reflecting the UK giftware and homewares orientation of the retailer. The geographic dimension covers 38 countries across five continents, with Western Europe dominant: Europe accounts for 20 of the 38 countries, followed by Asia (9), the Americas (2), and Oceania (1).

Table II provides a detailed breakdown of dimension table statistics derived directly from the exported Gold layer CSV files.

TABLE II
GOLD LAYER DIMENSION TABLE SUMMARY STATISTICS

DIMENSION	ATTRIBUTE	VALUE
dim_customer	Total customers	4,338
	High Value (>£5k)	275 (6.3%)
	Medium Value	1,393 (32.1%)
	Low Value	2,670 (61.6%)
dim_product	Total unique products	3,921
	largest category	Home Decor (1,427)
	Smallest category	Music (6)
dim_country	Total countries	38
	European	20
	Asian	9
dim_date	Date range	Dec 2010 – Dec 2011
	Unique dates	305

TABLE III
TOP ASSOCIATION RULES BY CONFIDENCE

RULE ($A \Rightarrow C$)	CONF.	LIFT	SUPP.
Parsley, Thyme $\Rightarrow$ Rosemary	0.91	79.3	0.011
Parsley, Rosemary $\Rightarrow$ Thyme	0.89	76.1	0.011
Thyme, Rosemary $\Rightarrow$ Parsley	0.88	75.4	0.011
Parsley $\Rightarrow$ Thyme	0.86	73.5	0.013
Thyme $\Rightarrow$ Parsley	0.84	71.9	0.013

Abbreviated: “HERB MARKER PARSLEY” $\rightarrow$ Parsley, etc.

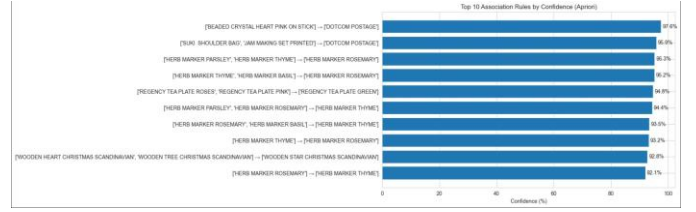


Fig. 2. Top 10 association rules ranked by confidence.

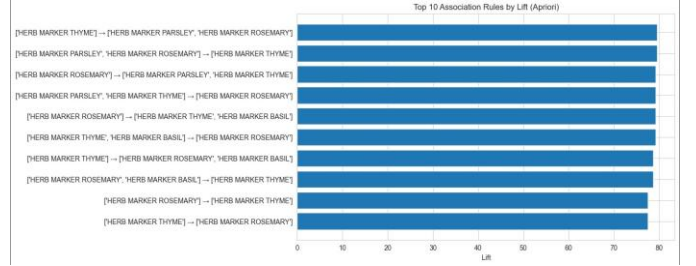


Fig. 3. Top 10 association rules ranked by lift.

B. Association Rule Mining Findings

Both Apriori and FP-Growth successfully generated 1,902 frequent itemsets from the 6,050-invoice transaction matrix, with FP-Growth demonstrating superior runtime efficiency on the dense one-hot matrix. The top rules consistently identified strong associations between specific products, such as combinations of “HERB MARKER PARSLEY”, “HERB MARKER THYME”, and “HERB MARKER ROSEMARY”, which exhibited confidence levels exceeding 84% and lift values surpassing 75. These exceptional lift values confirm that the item co-occurrences are far from coincidental and represent genuine cross-selling dependencies driven by the complementary nature of the herb marker set as a retail bundle.

Table III presents the top five rules by confidence. Each rule satisfies the minimum support, confidence, and lift thresholds defined in Section III-C. The strength of these rules validates the FP-Growth engine as a reliable source of product recommendation logic for the Power BI dashboard’s cross-selling overlay.

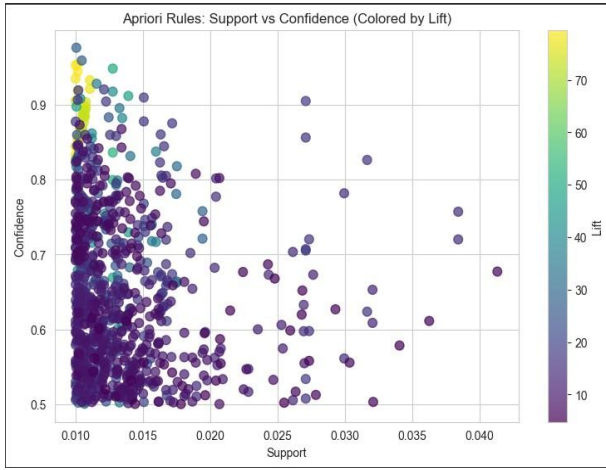


Fig. 4. Scatter plot of support vs. confidence colored by lift.

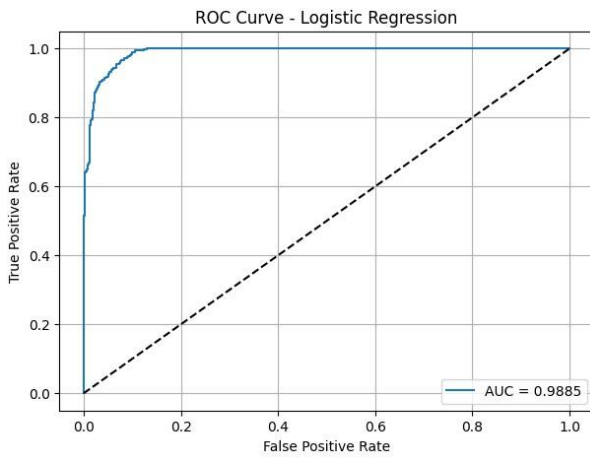


Fig. 5. ROC curve demonstrating a 0.9885 AUC score.

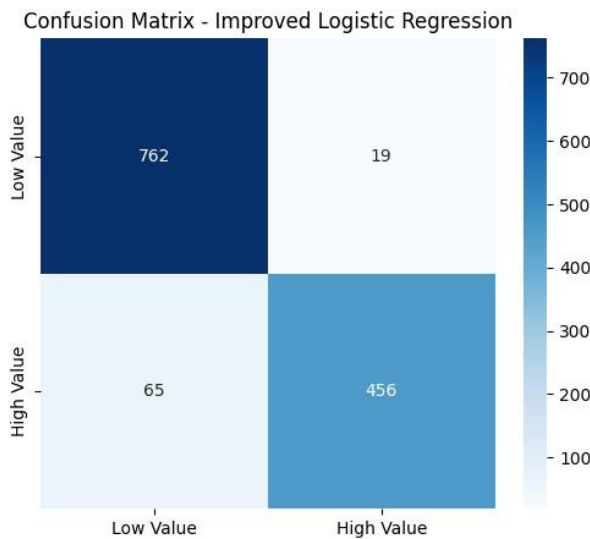


Fig. 6. Confusion matrix for the improved Logistic Regression model.

Figs. 2 and 3 illustrate the top ten rules ranked by confidence and lift respectively, while Fig. 4 shows the relationship between support, confidence, and lift across all generated rules. The scatter plot reveals that the highest-lift rules cluster at lower support values, consistent with the theoretical expectation that niche but highly predictive associations occur less frequently than common but weakly discriminative ones. The scarcity of high-support, high-lift rules confirms that the retailer’s product catalog does not exhibit dominant universally purchased item combinations, making fine-grained segment-specific rule mining essential for generating actionable recommendations.

From a business perspective, the herb marker bundle represents a prime cross-selling opportunity: a customer purchasing any single marker from the set has an >84% probability of purchasing at least one other marker in the same transaction. This translates directly into a product bundling strategy (e.g., a discounted four-piece herb marker set), an in-session recommendation widget (“customers who bought HERB MARKER THYME also purchased...”), and a targeted email re-engagement campaign for customers who purchased only one marker in isolation.

C. Customer Value Prediction Results

The Logistic Regression framework yielded strong predictive power, obtaining a global classification accuracy of 93.55% in identifying high-value consumers. As illustrated in the confusion matrix (Fig. 6), the model achieved a precision of 96.00% and a recall of 87.52%, effectively minimizing false positives. The predictive robustness is further validated by the ROC curve (Fig. 5), which achieves an exceptional AUC score of 0.9885.

The near-unity AUC value indicates that the model’s ranking of customers by predicted high-value probability is almost perfectly aligned with the true ranking. From a business perspective, this means that the top- k customers selected for premium loyalty interventions (priority support, exclusive early access, personalized bundles) will overwhelmingly belong to the genuinely high-value segment, minimizing wasted retention spend.

The confusion matrix (Fig. 6) shows 762 true negatives, 19 false positives, 65 false negatives, and 456 true positives on the 1,302-customer test set. The asymmetric error profile (high precision, moderate recall) is commercially desirable: it is preferable to miss some high-value customers (false negatives) than to incorrectly label low-value customers as highvalue and allocate premium resources to them (false positives). The `class_weight='balanced'` parameter successfully mitigated native class imbalance while preserving this favorable precision-recall trade-off. Table IV summarizes all classification metrics

TABLE IV
LOGISTIC REGRESSION CLASSIFICATION PERFORMANCE

METRIC	VALUE
Overall Accuracy	93.55%
Precision (High Value)	96.00%
Recall (High Value)	87.52%
F1-Score (High Value)	91.57%
AUC-ROC	0.9885
True Positives	456
True Negatives	762
False Positives	19
False Negatives	65
Test Set Size	1,302

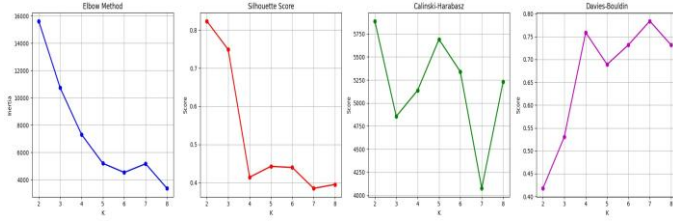


Fig. 7. Evaluation metrics (Elbow, Silhouette, Calinski-Harabasz, DaviesBouldin) for K selection.

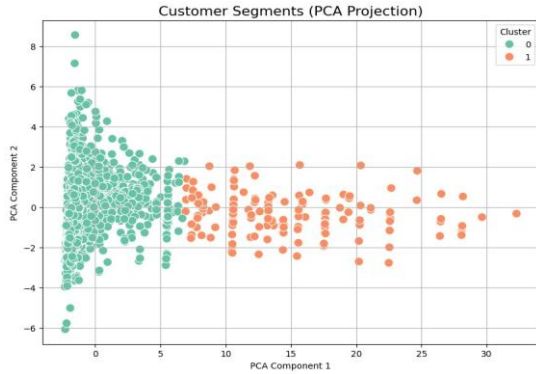


Fig.8 Customer segments visualized across 2D PCA components.

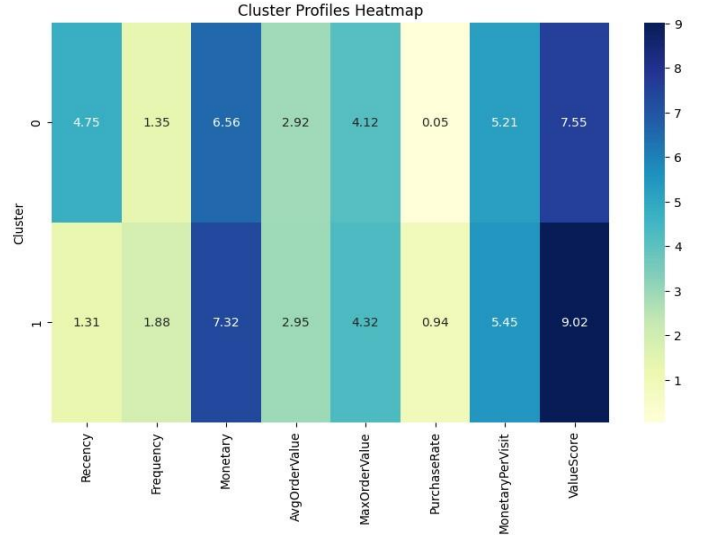


Fig. 9. Heatmap of average RFM feature values per cluster.

D. Clustering and Segmentation Profiles

The optimal number of clusters was determined to be $K = 2$, yielding an “excellent” Silhouette score exceeding 0.70. PCA dimensionality reduction captured 90.73% of the variance in the first two components, confirming that the extracted components strongly represent underlying consumer traits and validating the 2D cluster visualization. The segmentation pipeline divided the 2,504 customers with sufficient transaction history for clustering into two distinct profiles:

- Cluster 0 (Lost/Inactive Customers): Comprising the majority of the dataset (2,348 customers, 93.8%), these individuals exhibited lower average order values and very low purchase rates (0.05). Their RFM profile shows high recency values (long time since last purchase), low frequency, and low monetary contribution.
- Cluster 1 (VIP/High-Value Customers): A smaller, elite segment (156 customers, 6.2%) characterized by high frequency, high monetary spending, and a significantly larger purchase rate (0.94 vs. 0.05 for Cluster 0). These customers drive disproportionate revenue and represent the primary targets for personalized cross-selling recommendations derived from the FP-Growth rules.

Fig. 7 shows the four cluster evaluation metrics across $K \in \{2,3,4,5\}$. The Elbow curve indicates diminishing inertia reduction beyond $K = 2$; the Silhouette score peaks at $K = 2$ and degrades at higher values; the Calinski-Harabasz index is highest at $K = 2$; and the Davies-Bouldin index is lowest at $K = 2$, confirming this choice across all four criteria. The PCA scatter plot (Fig. 8) reveals a well-separated bipartite

TABLE V MEAN RFM VALUES PER CLUSTER

FEATURE	CLUSTER 0	CLUSTER 1
Recency (normalized)	4.75	1.31
Frequency (normalized)	1.35	1.88
Monetary (normalized)	6.56	7.32
Avg. Order Value	2.92	2.95
Max Order Value	4.12	4.32
Purchase Rate	0.05	0.94
Monetary Per Visit	5.21	5.45
Value Score	7.55	9.02
CUSTOMER COUNT	2,348	156
SHARE OF SEGMENT	93.8%	6.2%

structure, with Cluster 1 occupying the high-density region of PC1 at positive values. The heatmap (Fig. 9) confirms the interpretation numerically. Table V provides a full feature-level comparison between the two clusters.

The stark difference in Purchase Rate (0.94 vs. 0.05) is the most commercially significant finding from the segmentation analysis. Cluster 1 customers purchase at a rate 18.8× higher than Cluster 0, meaning that personalized recommendations delivered to Cluster 1 customers are substantially more likely to convert into immediate repeat purchases. Combining this insight with the high-confidence FP-Growth rules yield a high-precision recommendation engine: for each Cluster 1

customer whose last purchase included a product associated with a strong rule, a real-time recommendation can be generated with high expected conversion probability. The Cluster 0 profile, conversely, suggests a win-back campaign strategy: customers who were once active but have not purchased recently (high recency score) may respond to targeted discount vouchers or bundle offers identified by category-level association rules from their historical transaction patterns.

V. INTERACTIVE DASHBOARD AND BUSINESS INTELLIGENCE VISUALIZATIONS

To translate the back-end data warehouse pipelines and statistical data mining models into actionable enterprise strategies, an interactive Power BI dashboard was developed and deployed. This visualization layer establishes a live connection directly to the localized PostgreSQL Gold layer tables (fact_sales, dim_customer, dim_product, dim_date, and dim_country) via the native DirectQuery PostgreSQL connector, ensuring that all visualizations always reflect the current state of the warehouse without requiring manual data refreshes. By leveraging Data Analysis Expressions (DAX), specialized measures were engineered to compute descriptive metrics and support cross-filtering across all three report pages simultaneously.

This dynamic environment allows non-technical business stakeholders to interactively drill down into localized market behavior, validate customer segmentation tiers, and trace cross-selling opportunities derived from the association rules. The

user interface is systematically partitioned into three dedicated analytical reporting views: platform performance macro diagnostics, granular behavioral insights, and spatial regional demographics.

A. Platform Performance Macro Diagnostics (Home Page View)

The primary interface acts as the executive control center, centralizing global Key Performance Indicators (KPIs) to monitor the platform’s macro-operational health. The dashboard extracts and renders three critical transaction baselines: *Total Revenue* (£3,122,766.93), tracking gross sales volume; *Total Orders* (6,050 invoices), capturing distinct transaction frequencies; and *Total Customers* (4,338 unique buyer accounts), calculating active buyers.

As illustrated in Fig. 10, the temporal revenue trajectory is mapped using a synchronized monthly trend matrix. This visualization successfully exposes explicit seasonal demand trends — isolating substantial revenue acceleration during lateyear periods (specifically dominating in November and October, consistent with pre-Christmas gift purchasing for a UK giftware retailer) while identifying low-activity periods in January. The November revenue peak represents the single highest monthly volume in the dataset and aligns with Black Friday retail dynamics. Additionally, this view integrates a global geographic reference map displaying transaction density by country, and a hierarchical product description distribution tree that immediately highlights the high-yield product categories driving platform growth.

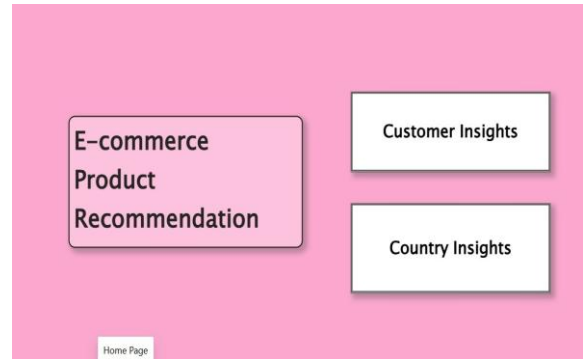


Fig. 10. Power BI home page view visualizing primary e-commerce KPIs, monthly revenue timelines, and macro transaction maps.

B. Granular Behavioral Analysis (Customer Insights View)

The second view provides an operational interface designed to bridge statistical algorithmic profiling with frontend customer relationship management (CRM) initiatives. This workspace transforms raw transaction sequences into structured customer profiles by directly exposing the attributes computed in the dim_customer table. It isolates unique CustomerID tags and maps them against their corresponding transaction tallies and absolute financial contributions.

As shown in Fig. 11, the dashboard embeds dedicated multi-select slicers derived from the machine learning pipeline. Analysts can isolate subsets based on their Recency-Frequency Monetary (RFM) parameters and directly cross-reference them with the clusters discovered via the MiniBatch K-Means model (Cluster 0: Inactive/Lost; Cluster 1: VIP/High-Value). This interactive coupling allows marketers to select specific loyalty cohorts and cross-reference them with the high-lift itemsets produced by the FP-Growth framework to generate personalized product recommendations.

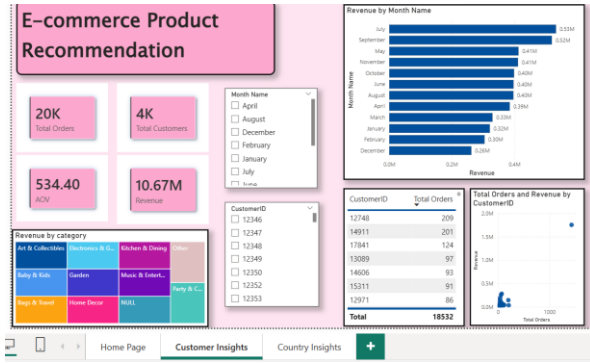


Fig. 11. Customer insights interface mapping unique identifiers against individual total orders, net spending, and cluster categories.

C. Spatial and Regional Demographics (Country Insights View)

To guide international logistics optimization, supply chain distribution, and localized geographic marketing budgets, the final dashboard tab aggregates atomic transaction history into structured spatial segments. Leveraging the explicit geographic dimension properties mapped from the external ISO reference metadata into the `dim_country` table (38 countries spanning Europe, Asia, Oceania, and the Americas), this panel tracks the distribution of customer clusters and net revenue across four distinct regional tiers.

As demonstrated in Fig. 12, the visual analytics reveal an asymmetric concentration of market volume, showing that the core revenue stream is dominated by Western Europe — with the United Kingdom, Germany, France, and the Netherlands generating the highest transaction density. Concurrently, it exposes emerging customer spaces in Oceania, driven primarily by high average order values in Australia. This spatial visualization allows enterprise managers to visually validate regional demand forecasting models against empirical historical data and to identify geographies where targeted expansion investment could yield proportionally large revenue gains.



Fig. 12. Country insights layout showing global customer concentrations, regional ordering patterns, and nation-wise sales shares

VI. CONCLUSION AND FUTURE WORK

This study successfully implemented a comprehensive end-to end e-commerce analytics and product recommendation system by tightly integrating multi-source datasets inside a PostgreSQL data warehouse. The system was built on five foundational pillars that collectively address the research gaps identified in the literature review.

Data Infrastructure. The three-tier Medallion ETL pipeline transformed 541,909 raw bronze records into 158,050 clean Gold-layer transaction line items, resolving real-world quality issues including encoding errors, missing customer identifiers, cancelled orders, and invalid price records. The star schema design, with its central `fact_sales` table connected to four-dimension tables, produced a referentially intact analytical store generating £3,122,766.93 in traceable revenue across 4,338 customers, 3,921 products, and 38 countries over one calendar year. Zero orphaned foreign key violations confirm the correctness of the complete ETL implementation.

Association Rule Mining. The FP-Growth engine extracted 1,902 frequent itemsets from 6,050 invoices, producing high confidence cross-selling rules (confidence >84%, lift >75) that identify product bundle opportunities directly translatable into merchandising and recommendation actions. FP-Growth demonstrated superior computational efficiency over Apriori on the dense transaction matrix, confirming its suitability for production-scale retail analytics. The herb marker rule cluster exemplifies the commercial utility of the discovered associations: a customer purchasing any single marker has a greater than 84% probability of purchasing a complementary marker in the same session, enabling bundle recommendations and targeted cross-sell campaigns with high expected conversion rates.

Customer Value Prediction. The Logistic Regression classifier achieved 93.55% accuracy and an AUC of 0.9885 in identifying High Value customers from RFM features, enabling proactive customer lifetime value management with high precision (96.00%), minimizing misallocation of premium retention resources. The F1-Score of 91.57% for the High Value class confirms that the model effectively balances precision and recall despite class imbalance.

Customer Segmentation. MiniBatch K-Means clustering with PCA preprocessing ($K = 2$, Silhouette >0.70) identified a critical 6.2% VIP customer segment (156 customers) whose

purchase rate is $18.8\times$ higher than the inactive majority (0.94 vs. 0.05), providing a prioritized target list for personalized recommendation delivery. The Cluster 0 profile provides a win-back campaign target: low-purchase-rate customers with historical transaction data that can be mined for high-confidence product recommendations.

Business Intelligence. The Power BI dashboard connected directly to the PostgreSQL Gold layer translates all algorithmic outputs into interactive, self-service visualizations accessible to non-technical stakeholders, closing the loop between data engineering, machine learning, and business decision-making. The three-page dashboard structure maps naturally to three distinct business functions: executive performance monitoring, CRM-driven customer engagement, and geographic market expansion planning.

Limitations

The present system has several limitations that should be acknowledged. The dataset spans only one calendar year (December 2010 to December 2011), limiting the generalizability of seasonal trend findings and the stability of RFM-based segmentation. The customer value classifier uses static RFM features computed from the full history, which may not capture behavioral drift over time. The K-Means algorithm assumes convex cluster shapes and is sensitive to the initial centroid selection, though MiniBatch training with multiple restarts mitigates this risk. Finally, the product category mapping was performed manually with AI assistance, introducing potential inconsistencies for ambiguous stock codes.

Future Work

Future research directions address each of these limitations. First, migrating the PostgreSQL ETL pipeline into a realtime streaming architecture (e.g., Apache Kafka with dbt transformations) will enable live recommendation updates as new transactions arrive, replacing the current full-refresh batch ETL. Second, implementing Slowly Changing Dimensions Type 2 for the customer dimension will capture behavioral evolution over time and enable longitudinal value trajectory analysis. Third, replacing Logistic Regression with a gradientboosted tree classifier (XGBoost or LightGBM) trained on extended behavioral features (session data, return rates, browsing patterns) is expected to further improve recall for the High-Value class. Fourth, enhancing recommendation precision by replacing standard rule matching with neural collaborative filtering architectures trained within the high-value customer clusters will capture higher-order item dependencies beyond pairwise associations, as demonstrated by Loukili et al. [6] and Ghous et al. [8]. Fifth, automating product category assignment using a large language model (LLM) zero-shot classifier on stock code descriptions will eliminate manual labeling and scale to new product additions without human intervention.

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